1. Apply Quick Sort algorithm to sort the number 8, 4, 2, 10, 7, 12, 1, 15
2. Apply Quick Sort algorithm to sort the number 5, 3, 1, 9, 8, 2, 4, 7
3. Apply Strassen's algorithm for matrix multiplication to multiply the following matrices and justify how the Strassen's algorithm is better

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

4. Solve Matrix multiplication of large integers using Divide & Conquer method A=1212 B=322
5. Explain Binary tree traversal algorithm with an example
6. With the help of State Space Tree, solve the 4-Queens problem by using Backtracking approach.
7. Apply Branch & Bound approach to the assignment to solve the instance of 0/1 Knapsack problem.
   n=4, m=10
   w = 4 7 5 3
   v = 40 42 25 12
8. Explain the class of P, NP and NP-Complete
9. Explain N-Queen problem with example.
10. What are decision trees? Explain how decision trees are used in sorting algorithms
11. Solve the greedy knapsack problem of capacity 5kgs. Items (1,2,3,4) Profit (5,9,4,8) Weight (1,3,2,2)
12. Solve the given instance of sum of subset problem s= {3,5,6,7} and d=15. Construct a state space tree
13. Explain Backtracking. List two advantages of backtracking strategy.
14. Apply Strassen's algorithm for matrix multiplication to multiply the following matrices and justify how the Strassen's algorithm is better

$$A = \begin{bmatrix} 4 & 5 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}$$